

Variable

```
In [ ]: Variables are used to store values.  
Python variables are only a name given to memory location
```

```
In [ ]: Python is dynamically typed language >>No need to assign data type of variable  
Everything is in python is an object.
```

```
In [3]: a=10  
type(a)
```

```
Out[3]: int
```

```
In [6]: b="Bhagyashri"  
print(type(b))  
c=4.5  
print(type(c))
```

```
<class 'str'>  
<class 'float'>
```

Rule to define Indetifiers

Indentifiers

```
In [ ]: Variable  
Class name  
function name
```

```
In [ ]: 1. A variables name must be start with alphabets (A-Z,a-z) or underscore(_)  
2.A variable name can only contain alpha numeric (a-z,A-Z,0-9)  
3.Variable name is case sensitive  
4.We can not use reserved keyword as a variable name.  
5.We can not use built in functions name as variable nae.
```

```
In [7]: name1="CodeWithbhagyashri"  
name1
```

```
Out[7]: 'CodeWithbhagyashri'
```

```
In [8]: name-5="Bhagyashri"
name-5
```

```
Cell In[8], line 1
    name-5="Bhagyashri"
    ^
```

SyntaxError: cannot assign to expression here. Maybe you meant '==' instead of '='?

```
In [10]: Player_name="Virat kohli"
Player_name2="Rohit Sharma"
print(Player_name)
print(Player_name2)
```

```
Virat kohli
Rohit Sharma
```

```
In [11]: _name="Python"
_name
```

```
Out[11]: 'Python'
```

```
In [13]: help("keywords")
```

Here is a list of the Python keywords. Enter any keyword to get more help.

False	class	from	or
None	continue	global	pass
True	def	if	raise
and	del	import	return
as	elif	in	try
assert	else	is	while
async	except	lambda	with
await	finally	nonlocal	yield
break	for	not	

```
In [ ]: print()
type()
help()
max()
min()
round()
```

Types of variables

```
In [ ]: 1.Local Variables>>We can access from inside the class
2. Global Variables>> We can access from inside as well as outside the class.
```

```
In [14]: a=10
b=20
c=30
def my_function():
    print("Value of variable a is:",a)
    print("Value of variable b is:",b)
    print("Value of variable c is:",c)
my_function()
```

Value of variable a is: 10
Value of variable b is: 20
Value of variable c is: 30

Global variable

```
In [15]: a=10
b=20
c=30
def my_function():
    print("Value of variable a is:",a)
    print("Value of variable b is:",b)
    print("Value of variable c is:",c)
my_function()

print(a)
print(b)
print(c)
```

Value of variable a is: 10
Value of variable b is: 20
Value of variable c is: 30
10
20
30

Local variable

```
In [18]: def my_function():
x=100
y=200
z=300
    print("Value of x is:",x)
    print("Value of y is :",y)
    print("Value of z is:",z)
my_function()
```

Value of x is: 100
Value of y is : 200
Value of z is: 300

```
In [19]: def my_function():
          x=100
          y=200
          z=300
          print("Value of x is:",x)
          print("Value of y is :",y)
          print("Value of z is:",z)
my_function()

print(x)
print(y)
print(z)
```

Value of x is: 100
Value of y is : 200
Value of z is: 300

```
-----
NameError                                Traceback (most recent call last)
Cell In[19], line 10
      7     print("Value of z is:",z)
      8 my_function()
----> 10 print(x)
      11 print(y)
      12 print(z)

NameError: name 'x' is not defined
```

```
In [24]: string="Data Science"

def addition(n1,n2):
    print("value of variable n1 is:",n1)
    print("value of variable n2 is:",n2)
    add=n1+n2
    print("Addition of n1 and n2 is:",add)
addition(3,4)
print(string)
#will get error because add is local variable
print(add)
```

value of variable n1 is: 3
value of variable n2 is: 4
Addition of n1 and n2 is: 7
Data Science

```
-----
NameError                                Traceback (most recent call last)
Cell In[24], line 10
      8 addition(3,4)
      9 print(string)
----> 10 print(add)

NameError: name 'add' is not defined
```

```
In [25]: string="data science"

def addition(n1,n2):
    print("value of variable n1 is:",n1)
    print("value of variable n2 is:",n2)
    add=n1+n2
    print("Addition of n1 and n2 is:",add)

def multiplication(n1,n2):
    print("Value of n1 is:",n1)
    print("Value of n2 is ",n2)
    mul=n1*n2
    print("Multiplication of n1 and n2=",mul)
addition(2,3)
multiplication(5,2)
```

```
value of variable n1 is: 2
value of variable n2 is: 3
Addition of n1 and n2 is: 5
Value of n1 is: 5
Value of n2 is  2
Multiplication of n1 and n2= 10
```

```
In [27]: string="data science"

def addition(n1,n2):
    print("value of variable n1 is:",n1)
    print("value of variable n2 is:",n2)
    add=n1+n2
    print("Addition of n1 and n2 is:",add)

def multiplication(n1,n2):
    print("Value of n1 is:",n1)
    print("Value of n2 is ",n2)
    mul=n1*n2
    print("Multiplication of n1 and n2=",mul)
addition(2,3)
multiplication(5,2)

print(string)
#will get error because add and mul are local variable
print(add)
print(mul)
```

```
value of variable n1 is: 2
value of variable n2 is: 3
Addition of n1 and n2 is: 5
Value of n1 is: 5
Value of n2 is  2
Multiplication of n1 and n2= 10
data science
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[27], line 18
     15 multiplication(5,2)
     17 print(string)
--> 18 print(add)
     19 print(mul)
```

NameError: name 'add' is not defined

Assigning values to the variables

```
In [28]: x=100
y=200
z=300
print(f"x=={x},y=={y},z=={z}")
```

```
x==100,y==200,z==300
```

```
In [29]: x=100,y=200,z=300
print(f"x=={x},y=={y},z=={z}")
```

```
Cell In[29], line 1
      x=100,y=200,z=300
      ^
```

SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?

```
In [30]: x,y,z=100
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[30], line 1
----> 1 x,y,z=100
```

TypeError: cannot unpack non-iterable int object

```
In [31]: x,y,z=100,200,300
```

Multiple values and Multiple variables

```
In [32]: x,y,z=100,200,300
```

```
In [33]: x,y=100,200,300
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[33], line 1
----> 1 x,y=100,200,300
```

ValueError: too many values to unpack (expected 2)

```
In [34]: x,y,z=100,200
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[34], line 1
----> 1 x,y,z=100,200
```

ValueError: not enough values to unpack (expected 3, got 2)

Multiple Values to single variable

```
In [35]: var1=100,200,300,400  
var2=(10,20,30,40)  
print(var1)  
print(var2)
```

```
(100, 200, 300, 400)  
(10, 20, 30, 40)
```

```
In [37]: x=var1[0]  
x
```

```
Out[37]: 100
```

```
In [39]: type(var2)
```

```
Out[39]: tuple
```

Data types

```
In [ ]: Numeric Data types:int,float,complex  
text data types:str  
sequence:list , tuple,range,set,frozenset  
mapping:dictionary  
Boolean:bool(True /False)(0 or 1)
```